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Co-design Workshops as a Step Towards Pilot Implementation for Complex Workplaces

A Case Study of London-Based Airport Future Workplace

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Abstract. This study has investigated how a complex multi-level organisation like a London-based airport can benefit from participatory design workshops using ecological interface design tools (i.e., Abstraction Hierarchy) for selecting or designing better future systems. Many complex organisations are using trials (pilot in our context) for selecting future autonomous technologies. This case is driven by a member of the innovation department of the airport and UX researchers. Our main objective was to employ participatory design and work domain analysis (WDA) as part of a framework to co-design and plan trials for future automated systems for smart work in airport terminal operations. The term automation in this paper also covers some of the so-called AI or more sophisticated automation. Over two weeks in two workshops in a London-based airport, we ran co-design workshops to help the decision-makers understand workplace needs and employee welfare while selecting future automated systems. We also explored potential issues in the work domain that the traditional user-centred design (UCD) methods could not systematically assess (e.g., information exchange or contextual effects). We conclude that WDA as part of co-design workshops prior to selecting the trials could be considered part of the pilot implementation for selection and design systems in complex workplaces, but it has complications.

Keywords: HCI · Participatory design · Automation · Personas · Interaction design · Abstraction hierarchy · Trials · Pilot implementation

1 Introduction

Automation and the introduction of Industry 4.0 interactive technologies in industrial work systems have brought new ambiguities in the challenges and burdens on interactive systems designers [15]. Socio-technical system design has identified and addressed several problems in understanding and developing complex autonomous systems [1, 3, 7, 9]. Therefore, service design and co-design processes are utilised to facilitate collaboration in new ways [18]. Designing future technologies needs more cross-organisational

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collaborations to produce innovative and creative outcomes. Many organisations prefer to select future systems by running trials and customising the existing systems instead of in-house designing. In such cases planning for a better selection of the trials is crucial and can bring more on-time competitive advantages. Trials can be defined as pilot implementation, and therefore their main objective is to learn. Hertzum et al. [13] defined pilot implementations as "an information-system development technique that aims to feed experiences from real use back into development by having users try out a system on a restricted scale before the design of the system is finalised". The pilot implementation can help organisations understand how to improve the system, adapt the organisation, and capture the benefits of introducing the system [13]. They can also help understand the less visible socio-technical issues and (hopefully) improve them in the subsequent intervention. As pilot implementations considered field tests, using the intended environment with real data, they can learn how a system may support its users in their work. Therefore, one of the potential improvements from the pilot is changes in UX. On the other hand, more studies on UX and innovation are needed in light of advances in Artificial Intelligence (AI) and the growing use of more sophisticated automation technologies [9]. Thus, using pilot implementation for these systems is beneficial in determining the value of the system and its design while reducing the implementation risks [13].

Simultaneously, there is a need for cooperation and better communication in human teams and individuals and AI systems to achieve better UX goals for future automation scenarios, namely AI/UX goals [1, 17]. Hence, future autonomous systems need to be carefully co-designed and tested in the field before or during the design. This should be properly engineered and bring some lessons learned to improve the system. In this paper, we report on a case study that employs participatory design and work domain analysis (WDA) [5] as a means for planning, co-design, and selecting trials for future automated systems considering AI/UX goals in the context of airport terminal operations.

This paper describes the process of a team of researchers and operational decision-makers of an airport in co-design two scenarios supporting smart work in the airport's terminals. Workshop sessions resulted in two future work scenarios leading to their rough prototyping using a WDA tool known as Abstraction Hierarchy (AH) [4]. The objective is to explore and illustrate how a London-based airport uses participatory design, interaction design, and WDA methods to decide on future autonomous systems trials based on the users' needs. The paper concludes with lessons we learned using such workshops prior to or as an initial stage to the pilot implementation phase.

2 Participatory Design and Work Domain Analysis

Approaches that have emerged to promote user engagement in system development have been implemented in different industries, such as aviation [19, 21, 22]. In addition, the participatory design approach has gained interest in the development of commercial and business applications [8]. The participatory design process emphasises mutual learning, as none of the participants, either 'designer' or 'user' knows everything.

WDA is a framework that supports and structures the analysis needed when designing a flexible and adaptive system [23]. Using WDA has two distinct advantages. First, WDA is a multi-dimensional analysis that incorporates the physical and the social environment

to provide a detailed description. Secondly, WDA can be paired with interface design [12] to generate new system designs.

From the WDA toolset, we decided to use AHs since they provide a discreet and complete description of a work domain at different abstraction and concrete levels. Furthermore, AHs are suitable to use by stakeholders from different backgrounds with little or no experience of WDA [2, 4]. AHs are used for human-machine systems design in empirical studies of operators' fault-finding strategies. AH determine what kinds of information should be displayed on the system interface and how the information should be arranged [12]. The main focus of AHs is to assess socio-technical relations, which allows the identification of contextual aspects otherwise only visible in pilot implementations, referred to as trials in our context.

3 Case Study Questions

This study followed a user-centric design approach for understanding domain experts necessities who are directly and indirectly working towards selecting the new technologies, their usability and UX goals. We offered a guideline to follow in the form of a co-design workshop based on our empirical for future systems. We co-created scenarios, personas, and AH as design methods to answer several questions: can co-design workshops lead to optimal design/selection of future autonomous systems? Can this be acceptable as part of the pilot implementation?

3.1 Workshop's Settings

We noticed that multi-levelled large organisations (i.e., our airport) mainly select their future technologies by running trials and observing how they proceed. However, customer and employee satisfaction and UX in the trials were not easily assessed and articulated or, in many cases, neglected. Our observations on the trials and the shortfalls seen regarding the use, usability, and UX of these tested technologies helped us suggest the airport user-centred design tools and methods for introducing new automated systems.

A success factor in any organisation is aligning the IT and business strategies, which is also true for socio-technical approaches. Thus, the strategy makers (managers) should be aligned with the system designers for successful implementation, including any pilot implementation. Otherwise, it will result in inefficiency in work, development and testing of the system, and failure in the pilot implementation. In our observations, we have noticed that one of the main factors neglected in selecting or running a trial is the primary users' (employees and end-users) need and requirement, which should assess technically and socially in the bigger organisational picture. Before selecting the future systems and replacing the new system with employees' day-to-day practices, strategy-makers should consider the culture, reward systems, employees' wellbeing and more socio-technical factors.

Considering these factors, we suggested that implementing co-design workshops with their domain expert (employees) before selecting future trials is beneficial for both

the strategy makers and the end-users. These types of selected trials are expected to have better user engagement and interaction than the others.

This suggestion triggered two co-design workshops with 14 participants, including researchers, industry decision-makers, and employees from different departments. Two researchers acted as facilitators in the workshops and used various topics and techniques to engage participants in prototyping system concepts. This system could be an autonomous system that is required in their departments. All participants had enough experience with autonomous technologies and leading Industry 4.0 implementation. In pair, they have discussed their proposed systems for short and concrete experiences in the future.

After a short cognitive walkthrough and group discussion for each system, the participants ranked the systems based on their value proposition and benefits for the airport. From a list of suggested scenarios (automated helpers, autonomous tugs, context-aware guest invites system, baggage tracing, and smart asset management.), participants selected two; Autonomous Tug and Pushback Taxi (ATPT) and Automated Asset Management and Maintenance (AAMM).

In workshop 2 we used Customer Job Canvas (part of value proposition canvas), persona's template, AI/UX goals [17] and AI guidelines [1], sketching, and AH [6] to prototype the short-listed future scenarios selected in the workshop 1. Researchers fulfilled facilitator roles supporting unfamiliar participants with the design tools while engaged as active designers in other tasks. Participants found it easier to work with the tools where they could have a role played in them, such as an employee or the end-user. Therefore, their engagement was high in customer job canvas, persona, sketching and testing AI/UX goals, compared to the difficulties faced while working on AH layers for the proposed systems. Thus, facilitators had to provide more tangible examples to shape the AH five layers; physical form, physical function, generalised function, abstract function and functional purpose. In developing AHs participants both analysed their work domain (e.g. airport service) and work tasks (e.g., checking people in).

In the last stage of workshop 2, each group walked through their concept [25] using their low fidelity prototypes and received feedback from other colleagues and decision-makers on how likely the idea was feasible. Finally, group members evaluated their design with UX goal templates [1, 17]. The quick result of this workshop was rejecting one of the trials due to functional viability and employees' welfare.

3.2 Data Collection and Analysis Methods

Data were gathered during the workshops via various sources. All group walkthroughs and final discussions were audio-recorded and transcribed verbatim. The data collected are audio recordings of the group work sessions during the workshop and written documents delivered during the workshop, such as personas, sketches, and observation notes. All in all, we analysed two scenarios' walkthroughs followed by the discussion and feedback from other teams lasting app 40 min, and we report the findings partially in this paper. Similarly to Friess [10], we analysed the conversational turns, but we left out single words and irrelevant conversations. The analysis started with open coding to provide a solid interpretive pattern and foundation for the thematic units [11]. For this paper, we only used some of the themes related to the pilot (trial) selection and

implementation and the benefits the decision-makers mentioned about the workshop, such as "UX goals", "employees' welfare", "financial viabilities", and "Risks of sharing responsibilities between the machine and human".

4 What We Learned - Pros and Cons of Such Workshops

We have learned that combining external academics and company employees can bring more efficiency to company workshops by providing recommendations viable on the industry that can be academically backed up. We also learned that employees are the best consultants for analysing the functions allocations for an optimal distribution of both functions and tasks between a partly autonomous system and the user [3]. Also, we observed that in complex jobs (e.g., terminal asset manager) in which employees deal with a high volume of data, there is more desire to share the work with a machine (e.g., robots and AI systems) rather than in low-skilled jobs (e.g., receptionist). This is specifically the case for tasks with a range of unpleasant, repetitive, too exhausting, or unsafe nature [16, 24]. We also learned that trust and safety considerations are more important in aviation than in other sectors as aviation insurance policies associate AI technologies with risks and uncertainty as these technologies take a degree of control of humans. We looked into the possibility of using such workshops as the initial stage of pilot implementation, for which we add the pros and cons here.

4.1 Pros and Cons of Co-design Workshops Prior to the Trials

One of the main strengths of this method was discovering the users' pains and gains in an early stage before selecting the system or trials using value proposition canvas. This helped the decision-maker understand the current situation better and realise the future gains or requirements well. Such practices provide temporary opportunities for the stakeholders to quickly assess how the proposed future system can fail in the organisation's environment and culture. In addition, understanding of the work domain and employees can shape into a shared vocabulary of technical and social requirements between the developers and decision-makers. Users can help in prototyping the technical configuration of the future systems using Abstraction Hierarchy.

Participatory workshops can bring more user engagement and empowerment (e.g., employees) by involving them in the design process. There is also an opportunity of changing the design requirements before the actual pilot implementation based on testing the usability goals. Therefore, we argued that such workshops should be an initial stage to the pilot implementation or the feasibility study for the pilot implementation.

One of the main cons for these workshops is the involvement of academics as facilitators to provide knowledge to the participants before the co-design workshop and during using AH to determine the technical configurations. This may not be accepted in some industries. Also, strong participants can influence the group decisions in such workshops and make them bias. Participatory workshops with WDA may not be accepted as pilot studies; in many cases they should rather be labelled and treated as feasibility studies.

4.2 Co-design Workshop Settings

All participants confirmed that using trials and writing lengthy reports is not optimal for planning and designing future scenarios. They found the workshop a quicker way to assess if the system works in the airport domain or not. However, on the other hand, bringing all these decision-makers to a co-design workshop for a couple of hours was not an easy practice. For a better result in the ideation process, providing a context, scenario, and better facilitation is required. Previous studies cover how the ideation process can be more comfortable and quicker for the participants [14, 20]. We observed that selecting the right design tool can better reflect the participants' knowledge. For example, participants found AH complicated to use without the design researchers' help. Adding more tangible elements [20], such as pre-structured cards and easy-to-use collaborative tools, can always be beneficial.

Moreover, there is a need to use data-driven design tools for the co-design workshops for future scenarios. Preparing the environment to ideate is another important factor; participants should understand the scenario, products, and the future system's domain and environment. The process provided the participants with an exemplary scenario to generate and manage their ideas. Otherwise, we must provide sufficient (internal/external) data for the context and persona creation. In line with the environment preparation and facilitator's instructions, planning for a scenario in advance for every co-design workshop for future systems is essential. This can help the participant access context and a road map to look forward and share more relevant information.

By the end of workshop 2, most participants understood how a future automation prototype could better fit their work environment through increased awareness of the work domain. Furthermore, they have agreed that there is a relation between modes of discovery, design improvements, interaction, and socio-spatial aspects. At the end of the session, one senior manager praised co-design and WDA methods to select and design new trials of future automation use cases and prototypes for the airport.

5 Conclusion

This case study's main contribution is about helping the airport co-design automation scenarios to incorporate workers' UX and work. During the design process in the workshops, we used WDA to contextualise the systems and assessed employees' acceptance of these systems. The presented case study's main aim is to explore and illustrate how a London-based airport uses participatory design, interaction design, and WDA methods to make decisions on trials for future autonomous systems that incorporate workers' experiences. We show how these co-design workshops and WDA design tools can yield early contextual socio-technical insights typically only found during pilot implementations. We empirically assessed our hypothesis on the proposed systems ATPT and AAMM scenarios in an airport domain based on customer profiles, personas, scenarios, AH, and a rough prototype to achieve this goal. We observed that this method could help designers or decision-makers foresee factors that may not be systematically assessed with other traditional UCD methods during the design process (e.g., information exchange or contextual effects). Producing the five layers for AH more efficiently with the help of academics and providing more tangible examples was one of the main

motivations in producing the final designs. After the workshop, it was highlighted in the senior participants' responses that using this method helps them better understand workers and the aviation sector's needs and perspectives for selecting the new autonomous systems, which was not part of their practices.

All in all, before conducting pilots in such large organisations, WDA in the form of co-design workshops can be used to select the available trials and their pilot implementation. The result can be used as an improvement in the intervention and design. WDA methods will be more effective when the key domain workers and stakeholders participate.

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